Nexus

BizIQ – AI-Powered SME Decision Intelligence Platform

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

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Project Name: BizIQ – AI-Powered SME Decision Intelligence Platform

Version: 1.0

1. Project Overview

- **Problem Statement:** Small and medium-sized business owners often rely on handwritten notes, raw spreadsheets and manual analysis to manage their operations. While tools exist for visualization, they fail to provide actionable decision support or incorporate external economic factors (e.g., inflation, supply disruptions). As a result, business decisions are reactive, inefficient, and prone to financial loss.
- **Target Domain:** AI-driven Business Intelligence and Decision Support Systems for SMEs, with emphasis on **retail and small-scale operations**.
- **Proposed Solution Summary:** BizIQ is a next-generation business intelligence platform that transforms raw business data into actionable insights and decisions. By integrating structured data (CSV/Excel) with AI-powered reasoning and scenario simulation, BizIQ enables business owners to make informed decisions regarding pricing, inventory, and strategy. The platform combines dashboards, AI chat, and financial simulation into a unified decision-making system.

2. Background & Business Objective

- **Background of the Problem:** Traditional tools such as spreadsheets and dashboards provide descriptive analytics but fail to guide users on what actions to take. SMEs often lack expertise in interpreting data, resulting in missed opportunities and inefficiencies. Moreover, external economic conditions are rarely incorporated into decision-making due to lack of accessible tools.
- **Importance of Solving This Issue:** This system enables SMEs to:

- Make informed decisions without technical expertise
- Reduce operational inefficiencies
- Improve profitability
- Simulate and prepare for future uncertainties
- **Strategic Fit / Impact:** BizIQ represents a shift from analytics → decision intelligence, where AI acts as a business advisor. It empowers SMEs with capabilities traditionally available only to large enterprises.

3. Product Purpose

3.1. Main Goal of the System: To provide SMEs with an AI-powered platform that not only visualizes data but also recommends actions and predicts outcomes.

3.2. Intended Users (Target Audience):

- Small retail business owners
- E-commerce sellers
- Entrepreneurs and startup founders
- Non-technical users managing business data

4. System Functionalities

4.1. Description: BizIQ is a web-based system that allows users to upload business data, analyze performance metrics, interact with an AI assistant, and simulate financial scenarios. The system integrates data processing, visualization, and AI reasoning into a single workflow.

4.2. Key Functionalities:

- **Data Ingestion Module**: Upload CSV or Excel files, temporary storage of uploaded data, and automatic parsing using CSV parsers and Excel processing libraries
- **Business Dashboard:** Displays key performance indicators (revenue, gross, and margin), growth trends, visualizations using charts (Recharts), and real-time updates after data processing.
- **AI Business Assistant:** Chat-based interface with streaming responses and understands uploaded business data context to provides insights and recommendations.
- **Financial Simulation Engine:** Allows "What If" analysis: Change pricing, adjust cost of goods (COGS), modify marketing spend, predicts impact on profit, revenue and margins.
- **Decision Recommendation Engine:** Detects anomalies and inefficiencies to generate actionable decisions for pricing adjustments, inventory optimization and cost reduction strategies.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system integrates Z AI GLM (via OpenAI-compatible SDK), enabling context-aware reasoning (understanding the business data), structured output generation (return the outcome in human language) and natural language interaction (allow users to ask why).

4.3.2. Prompting Strategy: We use a multi-step prompting approach where raw business data is first processed into structured metrics, and then the AI generates insights, recommendations, and explanations in separate stages. This allows us to separate calculation from reasoning, which improves accuracy, reduces hallucinations, and ensures every decision is clear, consistent, and explainable for the user.

4.3.3. Context & Input Handling: The system processes both structured and unstructured inputs in a controlled and optimized manner. Structured data (CSV/Excel) is first parsed and summarized into key business metrics (e.g., revenue, profit, product performance) before being sent to the AI model. Unstructured inputs, such as user queries or economic scenarios (in JSON), are combined with these metrics to form the context window.

To handle oversized inputs, the system applies:

- **Aggregation:** Summarizing large datasets into key statistics
- **Filtering:** Selecting top-performing and underperforming products (Top-N approach)

The system supports moderate dataset sizes (e.g., a few thousand rows). If inputs exceed the acceptable limit:

- Data is automatically **truncated or summarized**
- Only the most relevant information is passed to the model

This ensures efficient token usage, faster response time, and consistent output quality.

4.3.4. Fallback & Failure Behavior: The system includes multiple safeguards to handle unreliable or unusable AI outputs:

- **Retry Mechanism**: If the response is incomplete or unclear, the system automatically re-prompts the model with refined instructions
- **Validation Layer**: Outputs are checked for structure and relevance before displaying
- **Graceful Error Handling**: If the model fails, the UI displays a user-friendly message instead of breaking the experience
- **Fallback Logic**: Basic rule-based insights (e.g., profit trends, low-performing products) are shown if AI responses are unavailable

5. User Stories & Use Cases

- **User Stories:** As a business owner, I want to upload my data so I can understand my business performance.
- **Use Cases (Main Interactions):**

Dashboard Analysis Flow:
Upload data → process → display KPIs

AI Chat Flow:
User query → AI analyzes → response generated

Simulation Flow:
User modifies variables → system predicts outcome

Decision Flow:
System detects issue → generates recommendation

6. Features Included (Scope Definition)

- CSV/Excel data upload
- KPI dashboard
- AI chat assistant
- Financial simulation
- Decision recommendations

7. Features Not Included (Scope Control)

- Real-time API integrations

- Mobile application
- Multi-user accounts
- Advanced ML forecasting models
- Industry-specific customization

8. Assumptions & Constraints

- **LLM Cost Constraint:** The system is designed to maintain low inference costs per user session by minimizing token usage.
  1. **Estimated Cost**: Each user session is estimated to use ~1,000–2,500 tokens, depending on dataset size and number of interactions.
  2. **Cost Optimization Strategies**:
     - Data Summarization: Only aggregated metrics (not raw data) are sent to the model
     - Multi-step prompting: Avoids redundant large prompts
     - Context filtering (Top-N products): Reduces input size
     - Streaming responses: Improves perceived performance without increasing token usage
- **Technical Constraints:** The system depends on external AI APIs (e.g., GLM/OpenAI-compatible endpoints), making it sensitive to: API downtime, rate limits. Data processing is limited to CSV and Excel file formats and the system applies a static uploaded datasets (no real-time integrations).
- **Performance Constraints:** Response latency is influenced by the AI model processing time and network/API delays. At the same time, large datasets require preprocessing (parsing, aggregation) which may introduce slight delays before results are displayed. Therefore, to maintain responsiveness, the input size is limited via summarization and filtering while excessively large datasets are truncated

- **User Input:** The system requires structured input data (CSV/Excel format) and proper column formatting for accurate processing. Output quality is directly dependent on data completeness and data accuracy. The AI system requires explicit user prompts (no autonomous operation) and may generate probabilistic outputs that require user interpretation.

9. Risks & Questions Throughout Development

- **Design Similarity:** BizIQ mitigates this by grounding all AI outputs in user-specific structured business data (KPI summaries, product-level metrics, and trends) rather than generic prompt-only generation. Each response is conditioned on:
  - Dataset-specific statistical summaries (e.g., top/bottom products, margins, trends)
  - Session-specific context window (user's uploaded file only)
  - Multi-step prompting that forces decomposition of:
  - data interpretation
  - insight generation
  - recommendation formulation
- **Frame Stability:** Frame stability is handled at the frontend visualization layer, not by the AI model.
  - All dashboard components use a fixed responsive grid system (e.g., Recharts + CSS grid layout constraints)
  - Charts are rendered from normalized metric objects, not raw AI-generated structures
  - Output from AI is restricted to structured JSON schemas, ensuring UI consistency
  - If data exceeds layout capacity, the system applies:
    - pagination (tables)
    - aggregation (charts)
    - Top-N filtering (visual focus)

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- **Drawing Interpretation**: BizIQ does not rely on raw interpretation alone. Instead, it uses a data validation + fallback reasoning pipeline:
    - Pre-processing validation layer
        - Checks required schema fields (e.g., revenue, cost, units sold)
        - Flags missing or inconsistent data before AI processing
    - Data normalization stage
        - Converts raw CSV/Excel into standardized KPI format
        - Handles missing values using rule-based imputation or exclusion
    - AI confidence handling
        - If dataset quality is low, AI is instructed to:
        - explicitly state uncertainty
        - rely on aggregated trends only (not granular claims)
    - Fallback system
        - If interpretation fails, system provides:
        - rule-based insights (e.g., profit = revenue - cost)
        - basic trend summaries instead of advanced recommendations

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